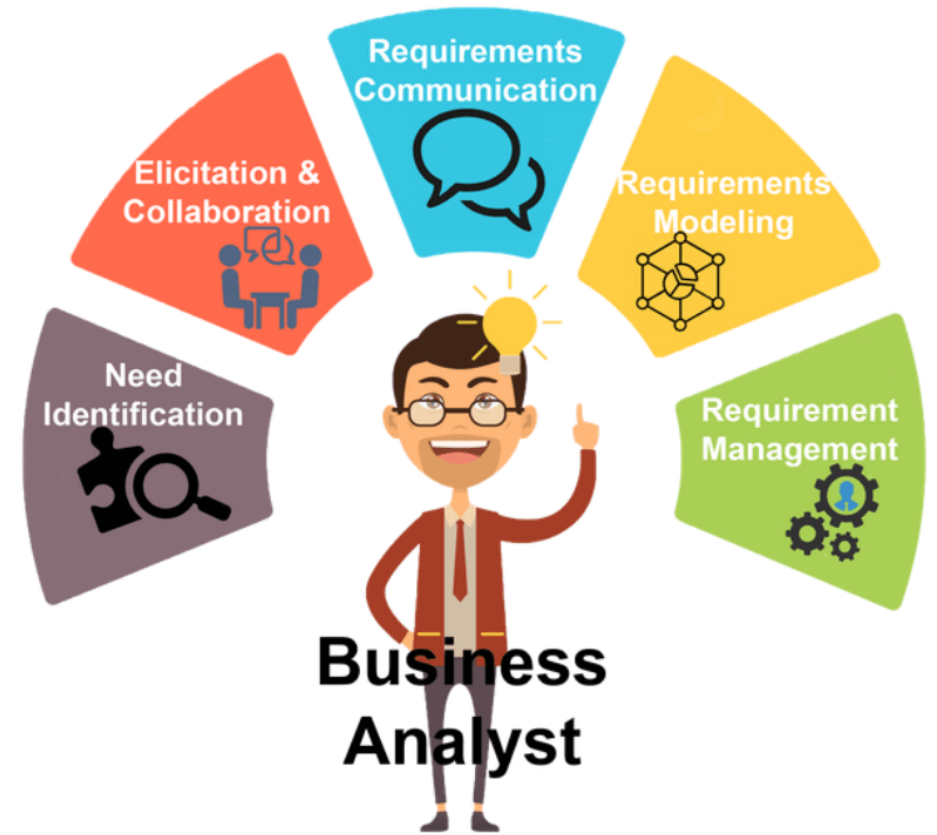


TrueCareer

Assessment-Driven Career
Guidance System

Business Analysis, Data Driven Recommendation,
and UI Prototype

TrueCareer



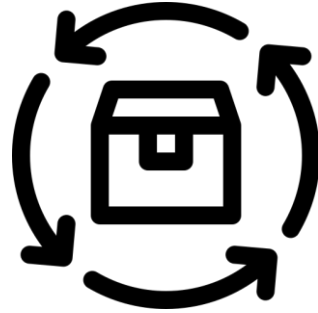
From Career Guidance Problem to Data-Driven Prototype

TrueCareer was developed to address a common challenge among Vietnamese high school students: many students struggle to choose suitable majors, understand career paths, and find reliable guidance.

TrueCareer combines:

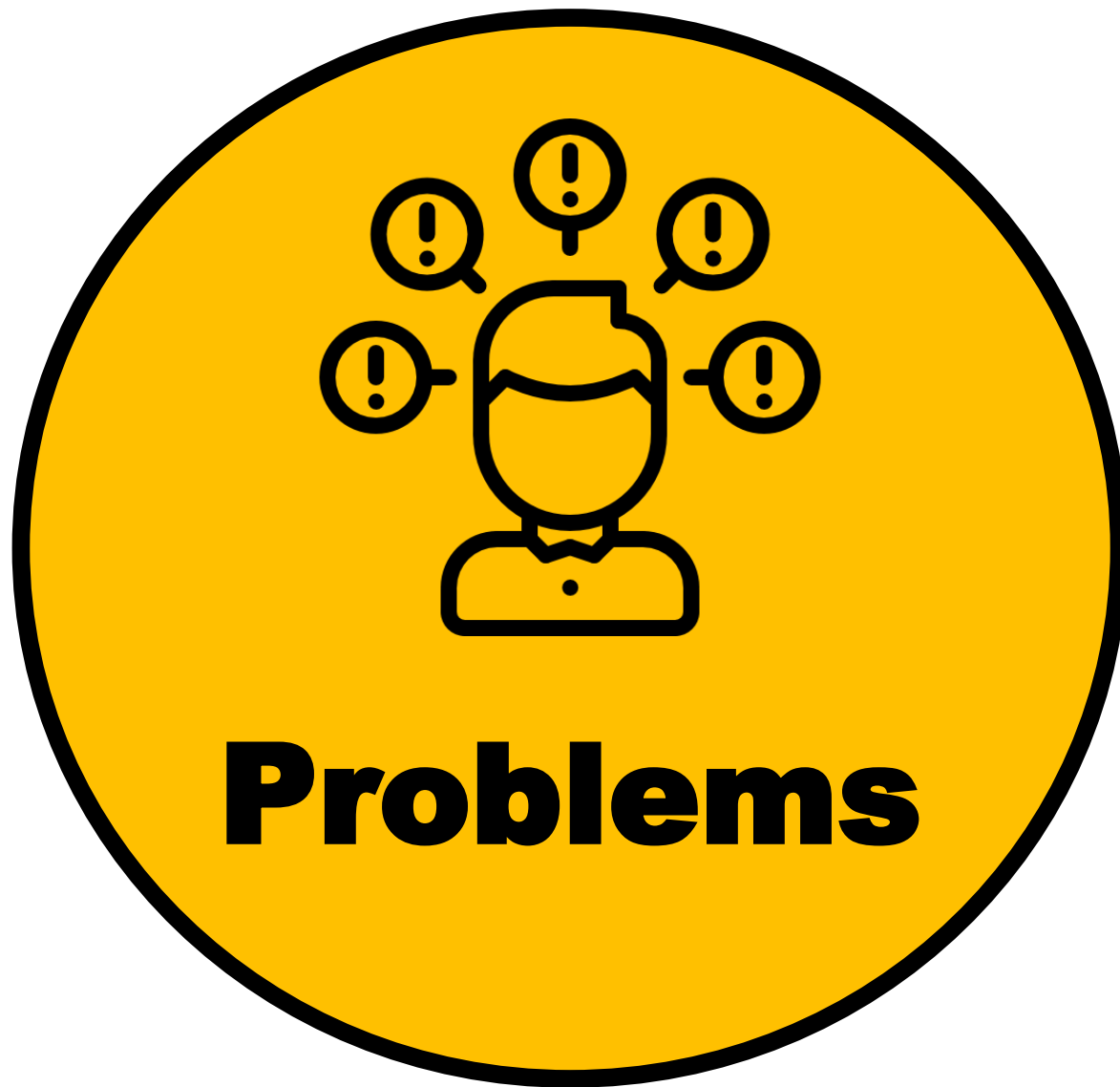
- Career assessment
- Student profile analysis
- K-Means student clustering
- Major and course recommendation
- Mentor matching
- Streamlit dashboard
- Axure UI prototype

Business Analyst Scope



Product Definition





Problems



11



12 Grade

Việt Nam



- Difficulty choosing suitable majors and career paths
- Career information is scattered across many sources
- Limited understanding of future labor market demand
- Limited time and experience for self-research and decision-making



Need strong professional background and 5+ years of experience.

Mentor recommendation should be based on student needs, not random selection

coaching/ mentor



**Goals &
Vision**

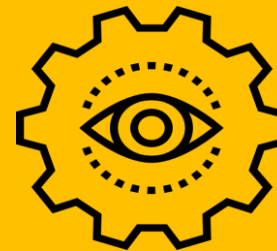
Goal & Vision



- Help Vietnamese high school students make better academic and career decisions through assessment-based guidance.



- Service packages
- Mentor consultation
- Future advertising



Vision

Become a data-driven career guidance platform that combines student assessment, recommendation logic, and mentor support.

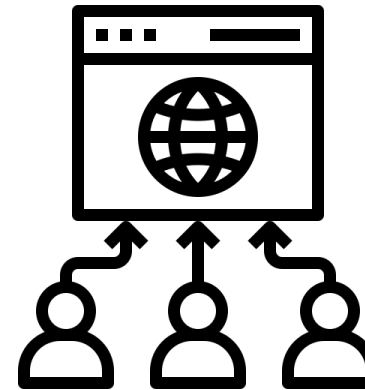


**Business
Objective**

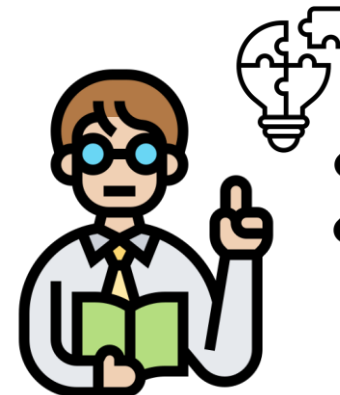
Business Objective



3 months



6 Months
200K



6 Months
> 70K



Business Cases

Business Cases



3-5

recommends majors,
courses, and career paths



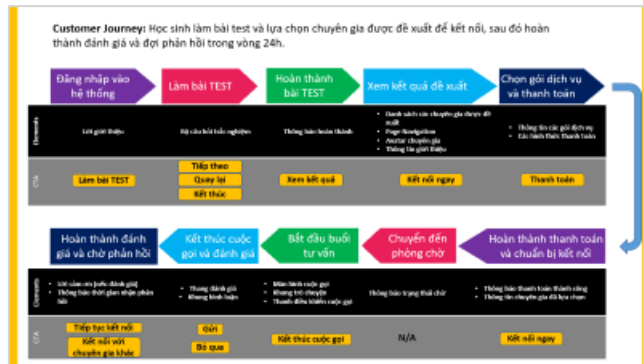
System suggests
suitable mentor type



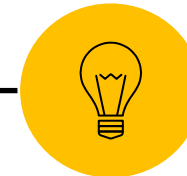
Raising income and
personal branding

Sử dụng lần lượt/ gói thanh toán

Customer Journey

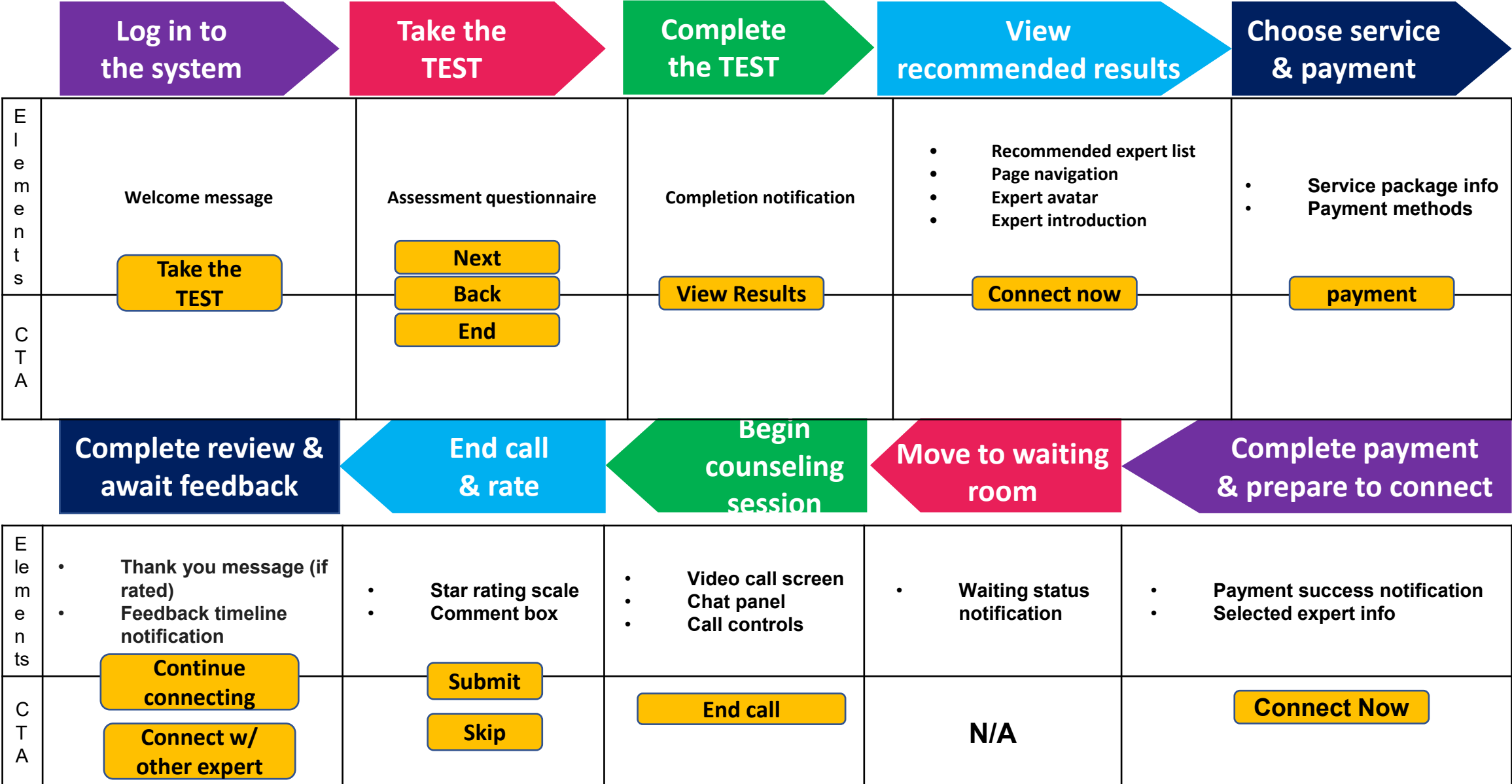


Customer Journey

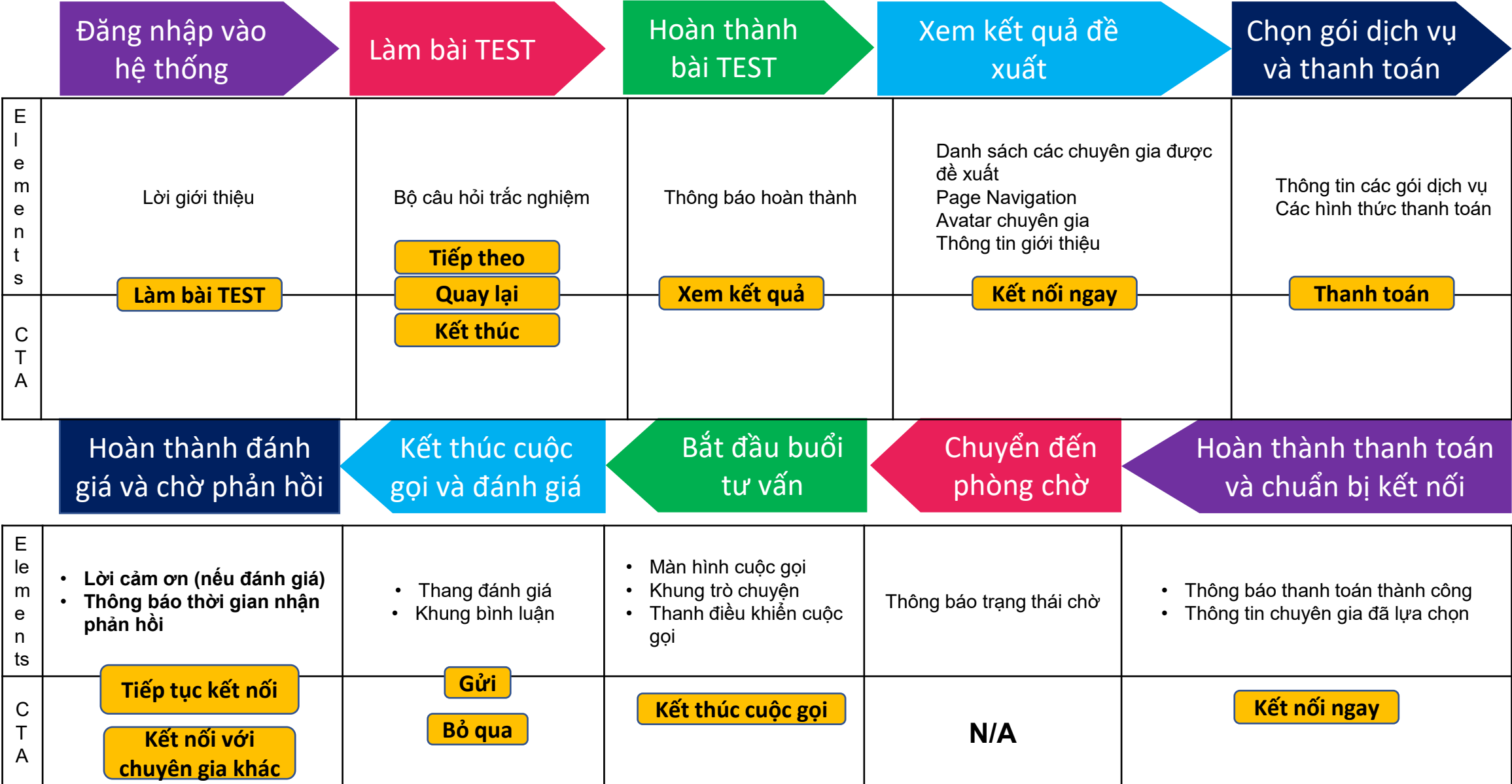


Card Sorting

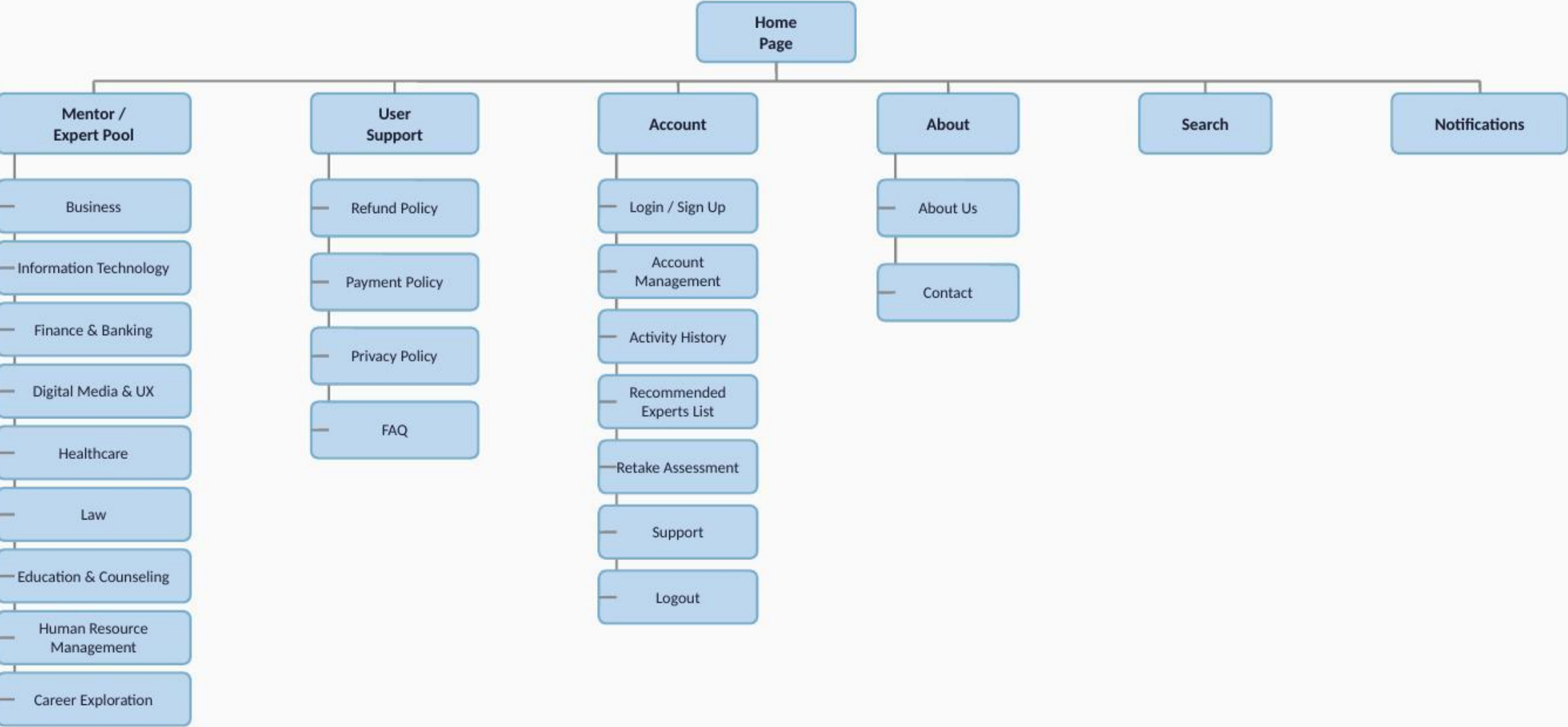
Customer Journey: Students complete the assessment, select a recommended expert to connect with, then submit a review and wait for feedback within 24 hours



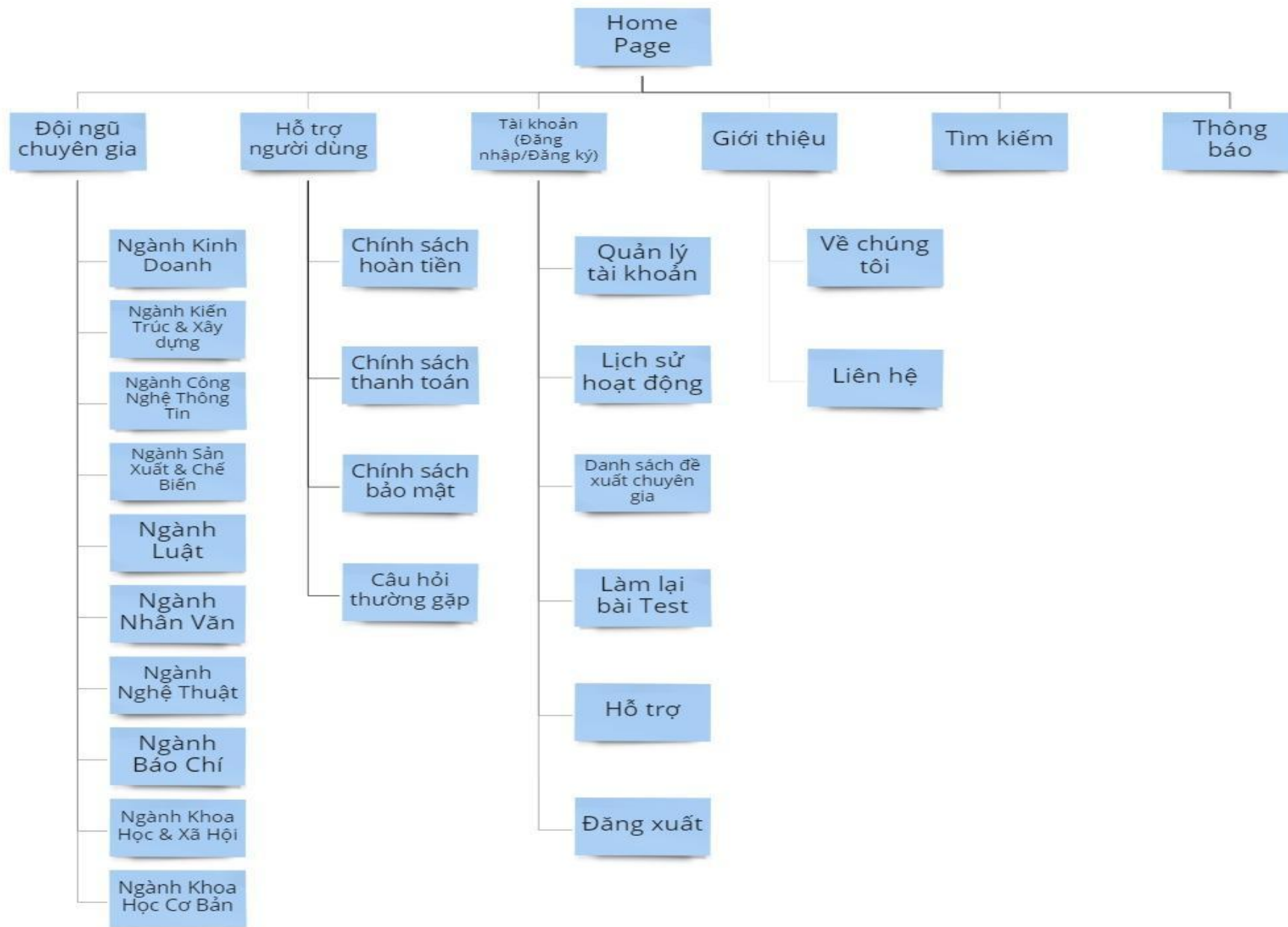
Customer Journey: Học sinh làm bài test và lựa chọn chuyên gia được đề xuất để kết nối, sau đó hoàn thành đánh giá và đợi phản hồi trong vòng 24h.



Card Sorting / Information Architecture



This structure shows how students navigate from the homepage to assessment, expert recommendations, account management, and support services.



System Solution



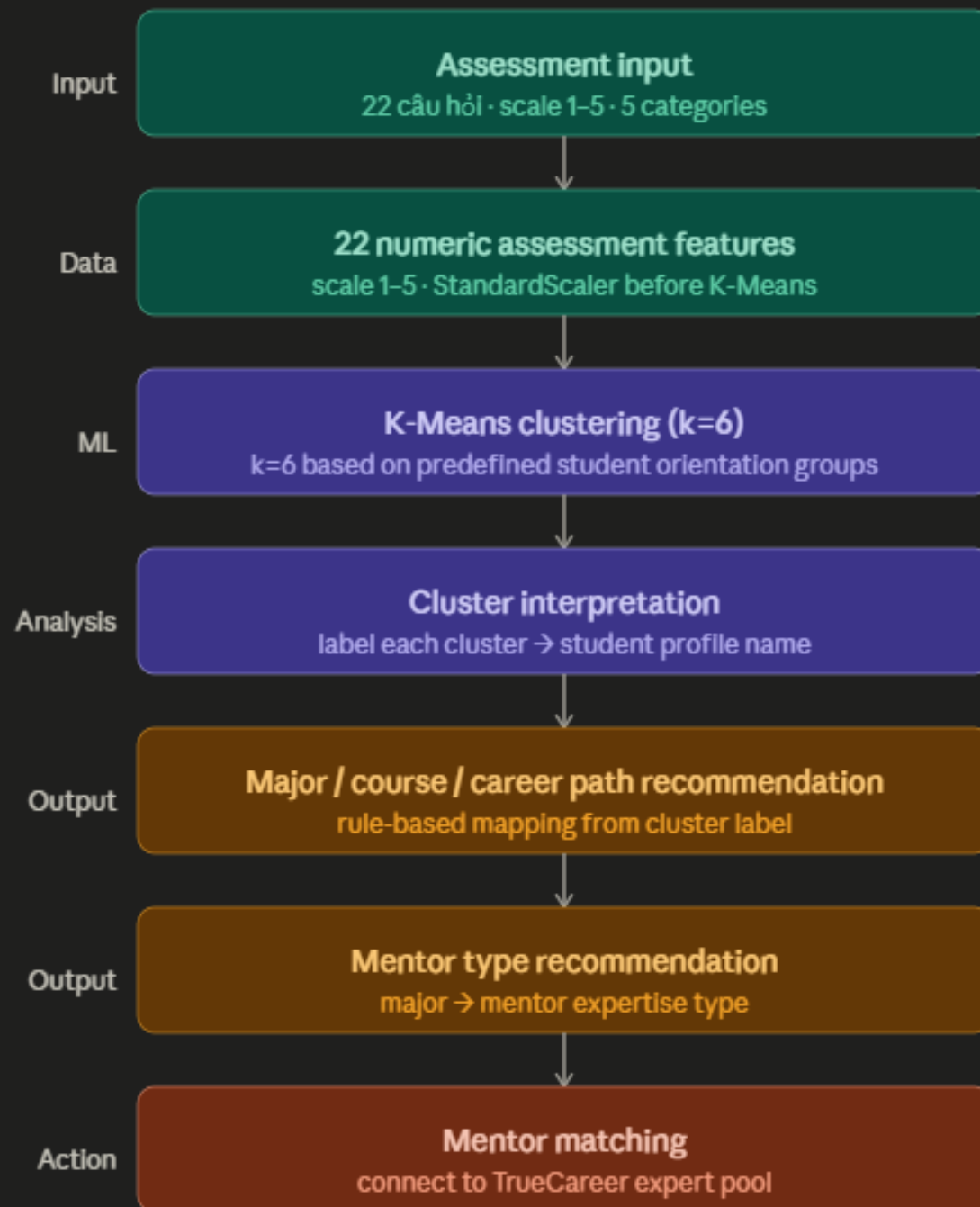
System Solution

TrueCareer solves the career guidance problem by connecting assessment data with recommendation logic.

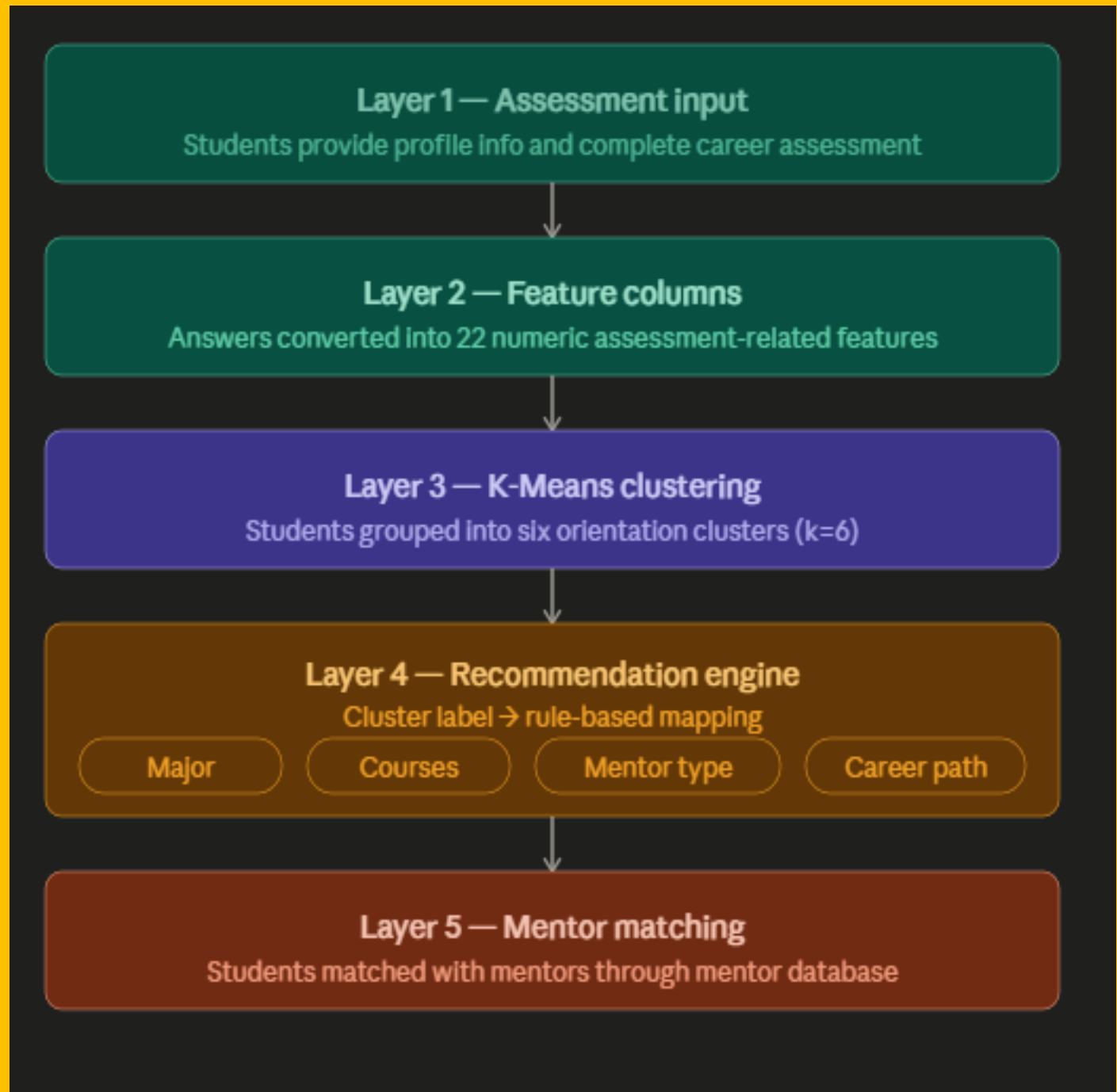
- Converts student assessment answers into structured feature scores
- Uses K-Means clustering to group students into six orientation types
- Recommends suitable majors, courses, mentor types, and career paths
- Matches students with mentors based on recommended mentor type

Data & Recommendation Logic

Key Point: K-Means is used for student clustering, while mentor matching is performed through rule-based mapping based on recommended_mentor_type.



System Architecture

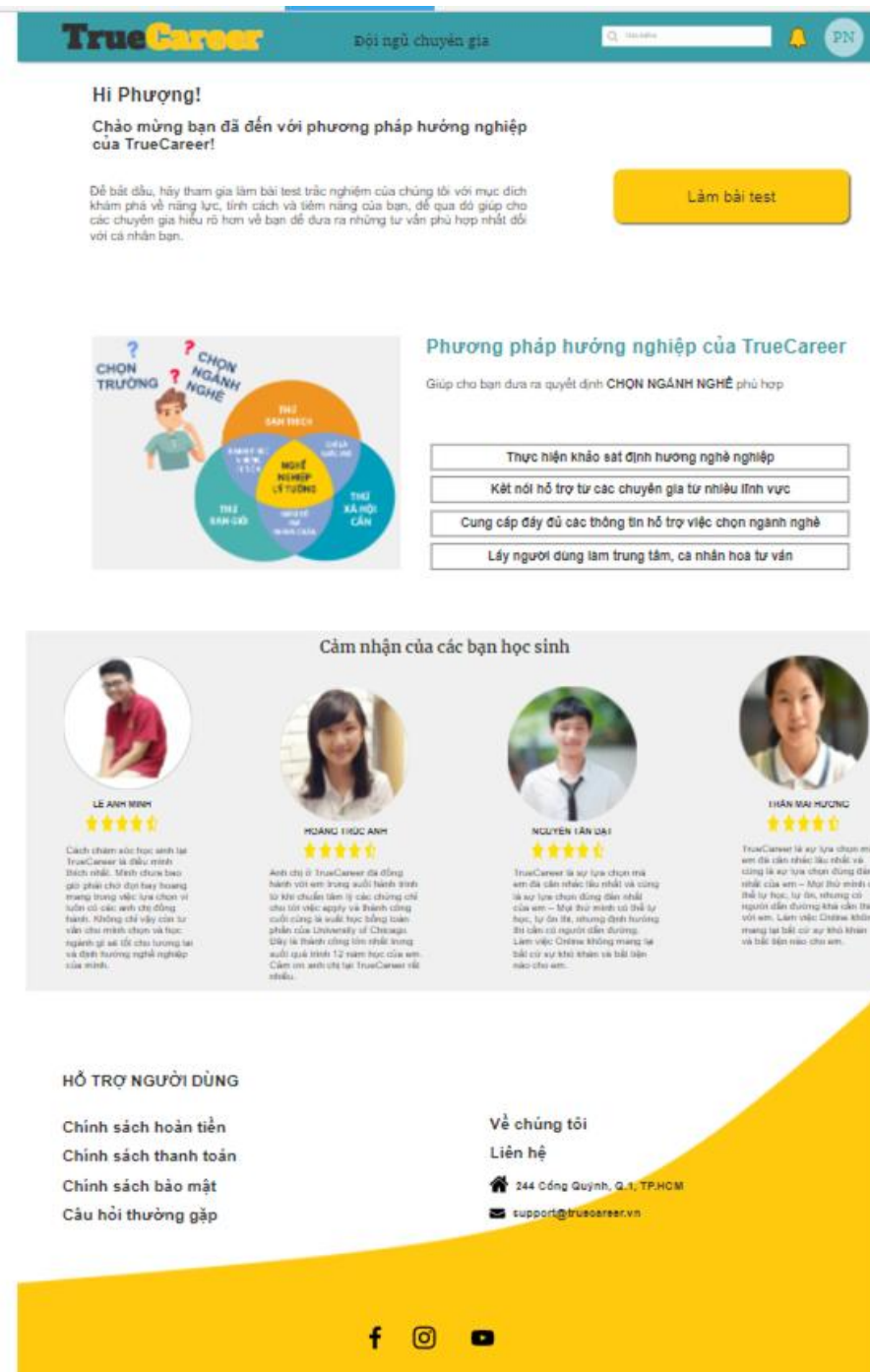


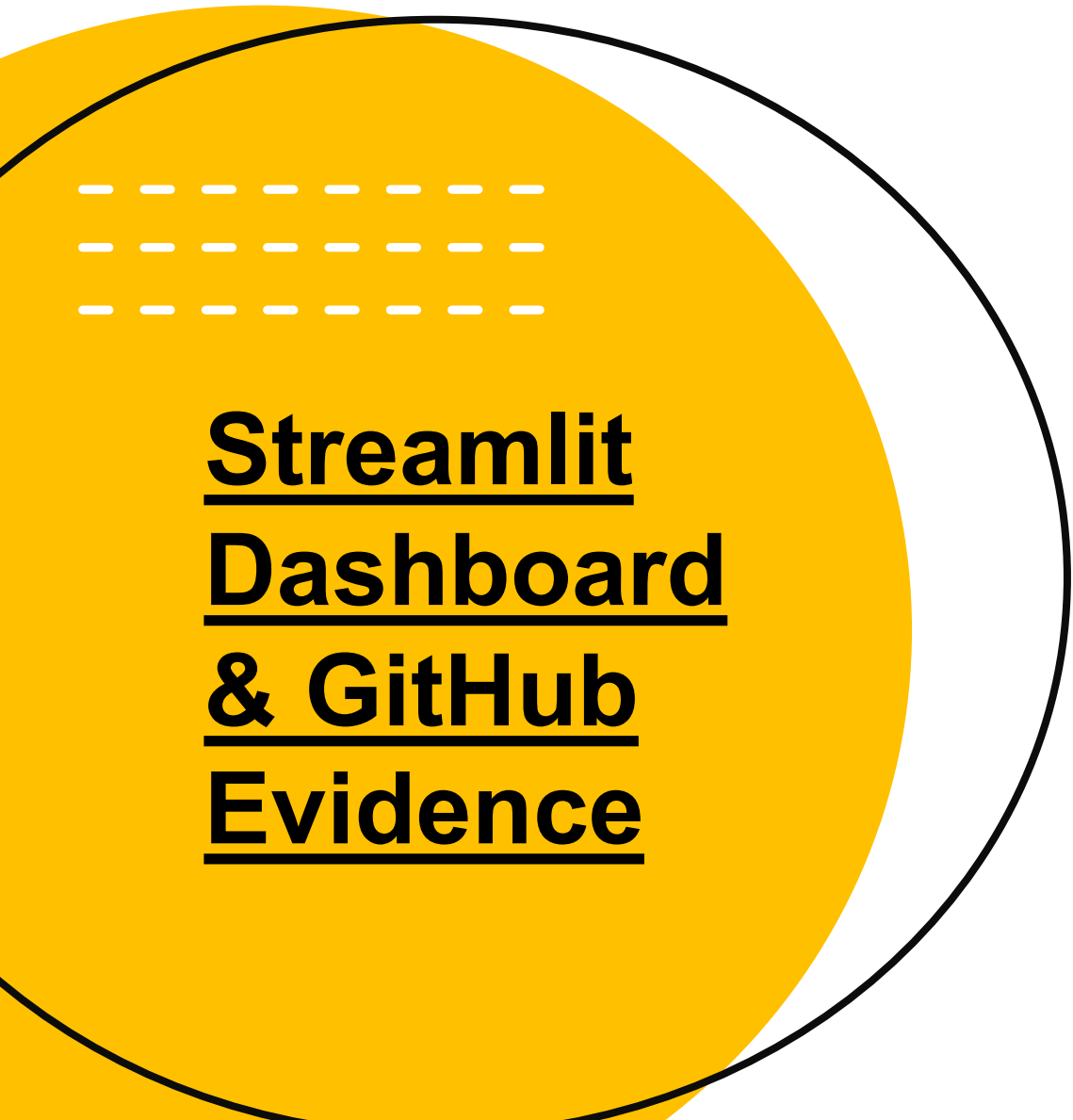
TrueCareer

Axure prototype demonstrates the user journey from assessment to mentor recommendation and consultation flow.

[Click Here](#)

Prototype





Streamlit Dashboard & GitHub Evidence

GitHub Repository Includes

Vietnam-context synthetic dataset

Python preprocessing script

K-Means clustering script

Recommendation summary script

Mentor database

Streamlit dashboard code

[Click Here](#)

Dashboard Demonstrates

Student cluster distribution

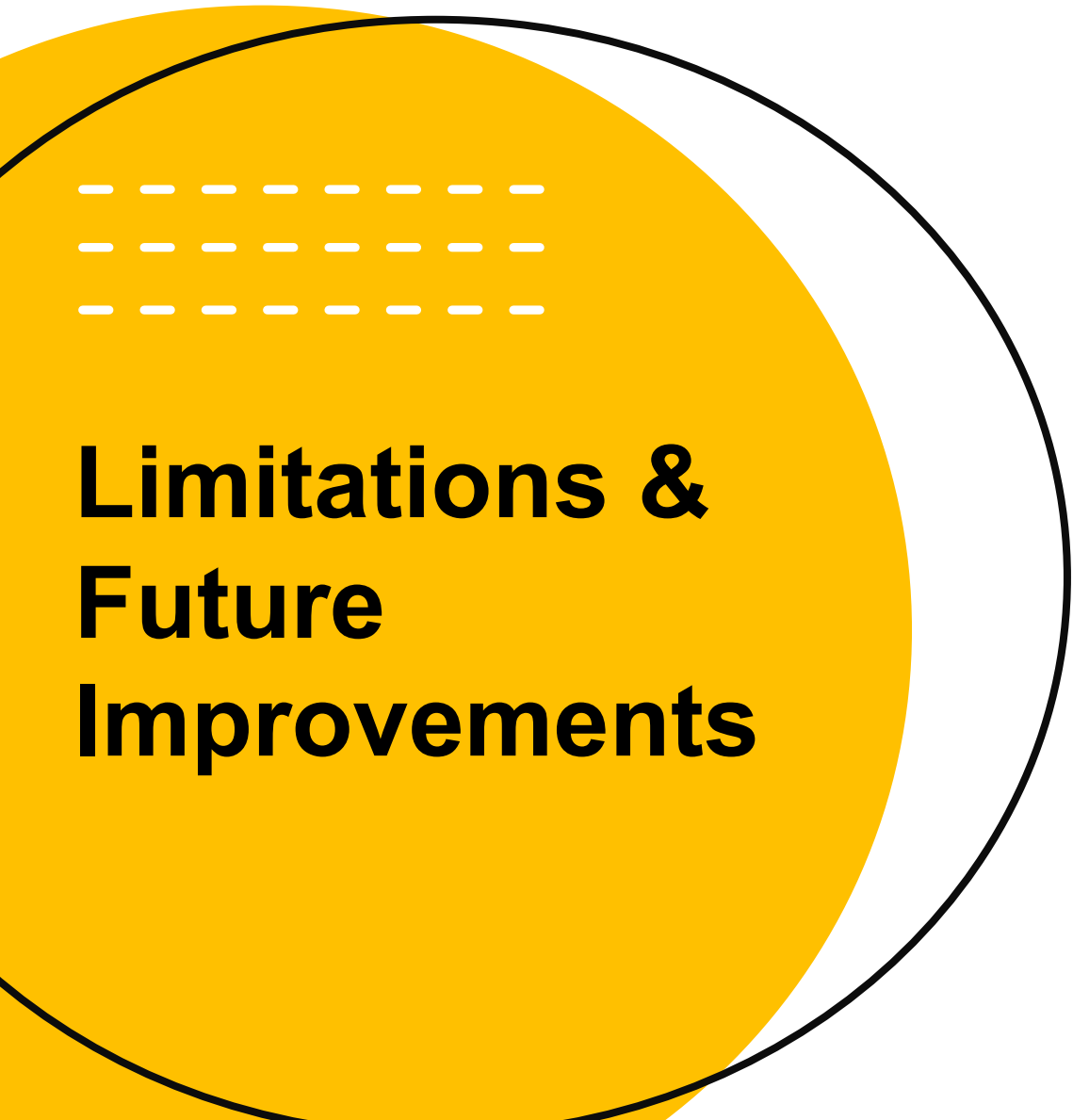
Recommended major distribution

Student profile filtering

Student-to-mentor matching

Mentor-to-student grouping

[Click Here](#)



Limitations & Future Improvements

Limitations

- Current dataset is synthetic
- K-Means clustering is exploratory
- Recommendation logic is rule-based
- Axure UI and backend are not fully integrated yet
- Real student feedback has not been collected

Future Improvements

- Collect real assessment data
- Add real-time assessment form
- Expand mentor database
- Improve recommendation accuracy
- Connect UI prototype with backend system

THANKS FOR LISTENING



ANY QUESTIONS

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